Ting-Chen Cho

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Education

Taipei Municipal Datong High School Graduate Jun 2021
National Taiwan University Expected Jan 2027

- **Bachelor of Science in Public Health**
- **Health Big Data Program**: Completed interdisciplinary course covering public health data analysis, big data applications, and statistical computing.

Certifications: NVIDIA Fundamentals of Deep Learning (2025)

Relevant Course

Data Analysis and ML with Python
Python Programming Foundation
Web Application Programming
Manufacturing Data Science
Epidemiology
Operating Systems
Database Management
Programming Design – C++
Linear Algebra
Computer Networks
Biostatistics I/II
Deep Reinforcement Learning

Experience

AirQuant – Alpha Team
Quant Developer

- Collaborated with the alpha team to design arbitrage strategies based on order flow and order book dynamics, contributing to the development of quantitative trading strategies.
- Supported strategy implementation in code, helping translate research ideas into testable trading algorithms.

Projects

SMART-D — LLM-Guided Multi-Agent Reinforcement Learning | *PyTorch, PPO, vLLM, Qwen2.5, Overcooked-AI*

- Built a cooperative multi-agent RL framework (PPO) that uses a frozen LLM as a training-time macro-planner generating high-level subgoals to accelerate exploration in sparse-reward tasks.
- Designed a single-network teacher/student controller with KL self-distillation and a coupled reward-shaping/distillation schedule, internalizing LLM guidance so the LLM is fully removed at deployment for zero inference latency.
- Implemented the state-to-text summarizer, LLM subgoal planner (JSON via vLLM-served Qwen2.5), and the Overcooked-AI evaluation environment; outperformed standard PPO across layouts and escaped coordination collapse on Forced Coordination.
- Source code available at: GitHub Repository

Medical Image Segmentation | *PyTorch, torchvision, SimCLR, TransUNet, UNet, wandb*

- Built a medical image binary-segmentation system with three architectures: UNet, TransUNet (ResNet50 + ViT), and SimCLR self-supervised pre-training.
- Used SimCLR contrastive learning to pre-train on unlabeled images, with a synchronized image/label augmentation pipeline.
- Trained with AMP, warmup scheduling, and early stopping, evaluated by Dice Score.
- Source code available at: GitHub Repository

Go Rank Prediction | *PyTorch, scikit-learn, CatBoost, NumPy, Pandas*

- Built a deep-learning ensemble predicting Go player ranks (1D–9D) by stacking Transformer, BiLSTM, and gradient-boosting models.

- Engineered features from game sequences and designed an ordinal-regression framework (CORAL loss) to handle rank class imbalance.
- Applied a multi-view inference strategy to improve stability on long game sequences.
- Source code available at: GitHub Repository

2D Generative Models — DDPM, DDIM, GAN, MeanFlow | *PyTorch, NumPy, Optimal Transport, wandb*

- Implemented four 2D generative model families from scratch in PyTorch: DDPM, DDIM, GAN, and MeanFlow.
- Built DDIM accelerated sampling and MeanFlow 1-step (1-NFE) generation to speed up inference over standard DDPM.
- Evaluated and compared models using Energy Distance and 2-Wasserstein distance via optimal transport.
- Source code available at: GitHub Repository

Bello — Social Meetup & Real-time Chat System | *Vue.js 3, Flask, PostgreSQL, JWT, Google OAuth, Vercel*

- Built a full-stack social platform for creating/joining physical meetups with real-time private and group chat, a friend system, and Google Maps location autocomplete.
- Designed a normalized PostgreSQL database (4NF) and a modular Flask backend covering auth, profiles, meetings, chat, friendship, and admin.
- Deployed on Vercel (Serverless backend + static frontend) with a Neon PostgreSQL cloud database.
- Source code available at: GitHub Repository

LINE Travel Bot — Conversational Travel Planner | *Next.js 16, TypeScript, PostgreSQL, Prisma, NextAuth, LINE Bot SDK, Gemini API, Tailwind CSS*

- Built a LINE-based conversational travel assistant that collects user preferences through multi-step dialogue and manages conversation state.
- Integrated Google Gemini API to generate personalized multi-day itineraries, with an admin dashboard (Google OAuth) for reviewing conversations and usage metrics.
- Designed a PostgreSQL schema with Prisma and deployed on Vercel with a Neon database and LINE webhook integration.
- Source code available at: GitHub Repository

Micropayment System — SSL/TLS Encrypted Client-Server Payment Platform | *C++, OpenSSL, POSIX Threads*

- Built a secure client–server micropayment platform (C++/OpenSSL) supporting account registration, login, balance query, and P2P transactions.
- Established SSL/TLS encrypted channels with OpenSSL certificate generation, exchanging and inspecting peer certificates between client and server.
- Designed a multithreaded server (thread pool + mutex) and a P2P model where each client forks a child process that listens for encrypted transactions from other clients.
- Source code available at: GitHub Repository

Leadership & Activities

2024 Harvard Taipei Academy Host Family Program

Volunteer

- Hosted visiting Harvard University students in Taiwan, introducing them to local Taiwanese culture and heritage sites.
- Planned and coordinated group meals, home cooking, cultural experiences, and outdoor activities to support cross-cultural exchange.

Freshman Orientation Camp

Orientation Group Leader

- Guided and supported incoming students during university orientation, facilitating group communication and coordinating team activities.

Rural Service Team

Volunteer

- Participated in a rural service program in Liujiao Township, Chiayi County, assisting with health education initiatives and conducting environmental surveys to support local community development.